

Supplementary Material: Extensive Ablation Study of Hamiltonian–Geometric Optimization

Reproducible computational supplement

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1 Scope and claims

This supplement reports raw multi-seed ablations generated by `run_ablation_study.py`. Paired initializations are retained across configurations. The exact full-matrix architecture is evaluated only where its parameter-dependent metric and finite-difference forces are tractable. The CUDA study evaluates the documented diagonal reduction and is labeled accordingly. No missing GPU run is presented as completed.

2 Experimental design

The architecture experiment crosses geometric force (G), memory force (M), and spectral-entropy force (S) in a complete 2^3 factorial under a curvature-derived Hessian metric and an unrelated positive-definite control metric. The chaotic quantum experiment repeats the same eight configurations with paired problem seeds and adds AdamW, heavy-ball, and entropy-descent references. Medians and nonparametric bootstrap 95% intervals use raw per-seed values. Diverged trials remain in the raw CSV and are counted. The completed data contain 40 architecture seeds at 300 steps, 10 quantum seeds at 40 iterations, and 10 CUDA seeds at 600 minibatch steps.

3 Full-architecture factorial results

Metric	Config.	n	Median loss	95% CI	Diverged
control	G0M0S0	40	-25.24	[-33.2, -19.39]	0
control	G0M0S1	40	-25.24	[-33.2, -19.39]	0
control	G0M1S0	40	-25.29	[-33.2, -19.39]	0
control	G0M1S1	40	-25.29	[-33.2, -19.39]	0
control	G1M0S0	40	-25.24	[-33.2, -19.39]	0
control	G1M0S1	40	-25.24	[-33.2, -19.39]	0
control	G1M1S0	40	-25.29	[-33.2, -19.39]	0
control	G1M1S1	40	-25.29	[-33.2, -19.39]	0
hessian	G0M0S0	40	-33.2	[-33.2, -25.29]	0
hessian	G0M0S1	40	-33.2	[-33.2, -25.29]	0
hessian	G0M1S0	40	-33.2	[-33.2, -25.29]	0
hessian	G0M1S1	40	-33.2	[-33.2, -25.29]	0
hessian	G1M0S0	40	-33.2	[-33.2, -25.29]	0
hessian	G1M0S1	40	-33.2	[-33.2, -25.29]	0
hessian	G1M1S0	40	-33.2	[-33.2, -25.29]	0
hessian	G1M1S1	40	-33.2	[-33.2, -25.29]	0

Table 1: Paired multi-seed full-architecture results.

4 Factorial effect estimates

Effects are twice the ± 1 -coded OLS coefficient on $\log_{10}(L - L_{\min} + 10^{-10})$. They measure association within this controlled design, not universal optimizer superiority. Paired t tests, Wilcoxon signed-rank tests, stan-

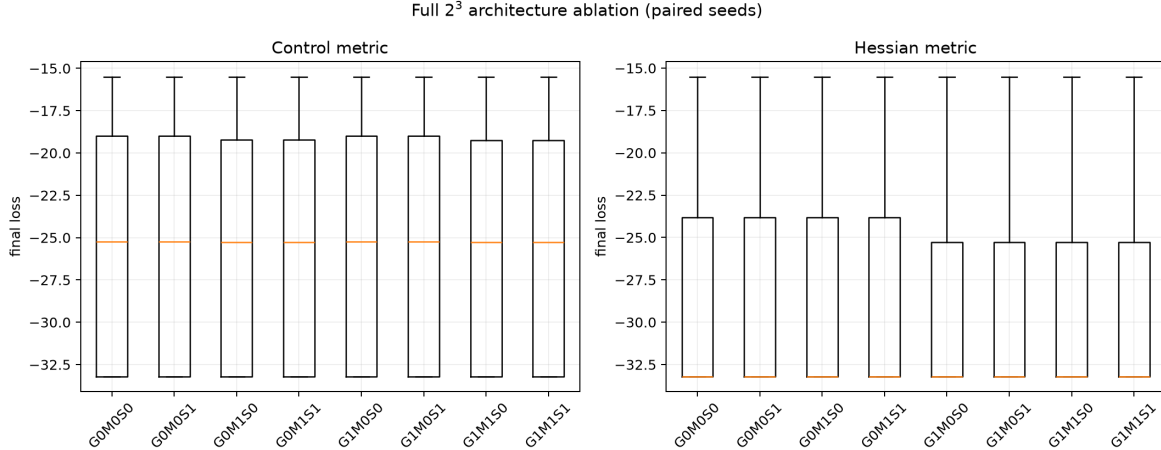


Figure 1: Final-loss distributions for the complete factorial.

Study/metric	Factor	Effect	<i>p</i>
architecture/control	G	-0.00527	0.991
architecture/control	M	-0.485	0.304
architecture/control	S	0.585	0.216
architecture/hessian	G	-0.321	0.541
architecture/hessian	M	0.189	0.719
architecture/hessian	S	2.08	9.25e-05
quantum/quantum	G	0.88	0.000157
quantum/quantum	M	0.305	0.171
quantum/quantum	S	0.222	0.318

Table 2: Main factorial effects. Full interactions and standard errors are in `results/factorial_effects.csv`.

standardized paired effect sizes, and within-family Holm corrections are reported in `results/paired_tests.csv` (29 prespecified comparisons).

5 Chaotic quantum-control replication

6 CUDA neural scaling ablation

This experiment uses the repository’s diagonal-metric Torch reduction; it is not evidence that a full Hessian or finite-difference geometric force scales to neural networks. CUDA execution was verified on **NVIDIA GeForce GTX 1050**. The protocol holds out the final 4096 examples, resets the minibatch stream for every optimizer within seed, and evaluates every 25 updates. Failures to reach validation accuracy 0.90 are right-censored at the final update.

7 Reproducibility and limitations

Software and device metadata are stored in `results/metadata.json`. Raw observations and histories are retained as CSV files. Hyperparameters are fixed before the seed sweep, so these results test robustness of those settings rather than granting each configuration a different tuning budget. The quartic task isolates mechanisms but cannot establish application-wide generalization. The quantum workload is small, and the CUDA benchmark uses a diagonal approximation because a dense metric is infeasible at neural-network

Configuration	n	Fidelity	95% CI	Runtime (s)
G0M0S0	10	0.9736	[0.9455, 0.9885]	3.878
G0M0S1	10	0.9765	[0.9257, 0.9856]	39.830
G0M1S0	10	0.9565	[0.9309, 0.9729]	3.853
G0M1S1	10	0.9600	[0.9192, 0.9785]	38.888
G1M0S0	10	0.8812	[0.8307, 0.9150]	39.302
G1M0S1	10	0.8565	[0.8206, 0.9162]	73.708
G1M1S0	10	0.8753	[0.8370, 0.9137]	40.299
G1M1S1	10	0.8748	[0.8190, 0.9163]	77.712
adamw	10	0.9861	[0.9764, 0.9891]	1.202
entropy_descent	10	0.9876	[0.9793, 0.9902]	2.341
heavy_ball	10	0.9930	[0.9903, 0.9953]	1.152

Table 3: Best-seen fidelity across paired kicked-top problem seeds.

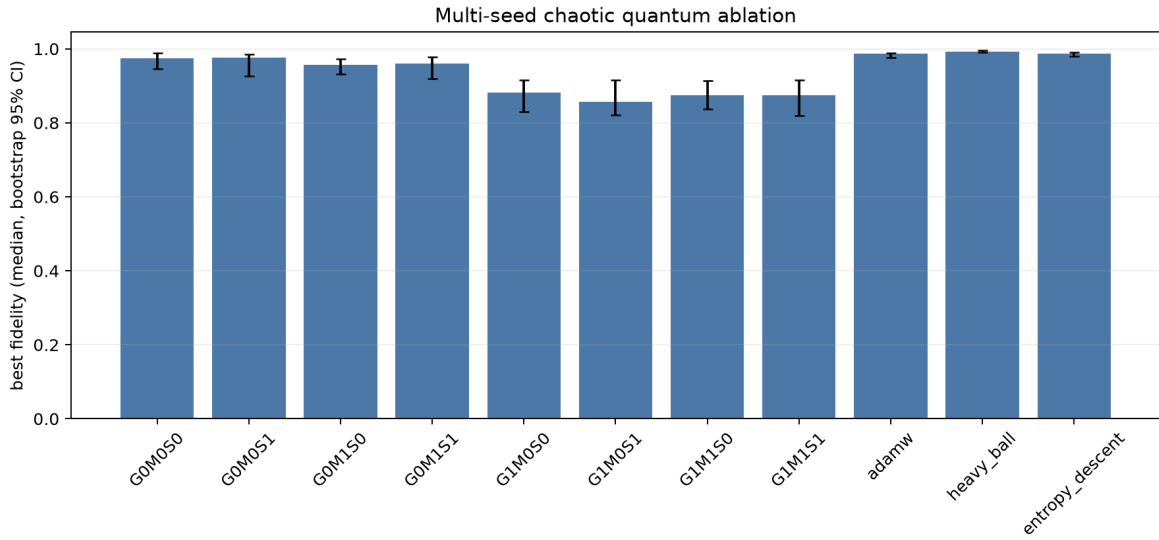


Figure 2: Median quantum fidelity with bootstrap confidence intervals.

scale. Claims should therefore remain specific to the measured tasks.

Configuration	n	Accuracy	95% CI	Runtime (s)
AdamW	10	0.8883	[0.8834, 0.8912]	3.27
HG_metric_momentum	10	0.8605	[0.8582, 0.8627]	5.74
HG_metric_no_momentum	10	0.9072	[0.9026, 0.9097]	5.66
HG_scalar_momentum	10	0.8628	[0.8569, 0.8663]	5.66
HG_scalar_no_momentum	10	0.9081	[0.9030, 0.9097]	5.70
SGD_momentum	10	0.9099	[0.9069, 0.9121]	2.67

Table 4: CUDA component-ablation results.

Configuration	Target hits	Updates [†]	Time (s) [†]
AdamW	2/10	450.0	2.50
HG_metric_momentum	0/10	—	—
HG_metric_no_momentum	9/10	461.1	4.43
HG_scalar_momentum	0/10	—	—
HG_scalar_no_momentum	9/10	461.1	4.46
SGD_momentum	10/10	345.0	1.56

Table 5: CUDA time-to-target; [†]means among successful trials.

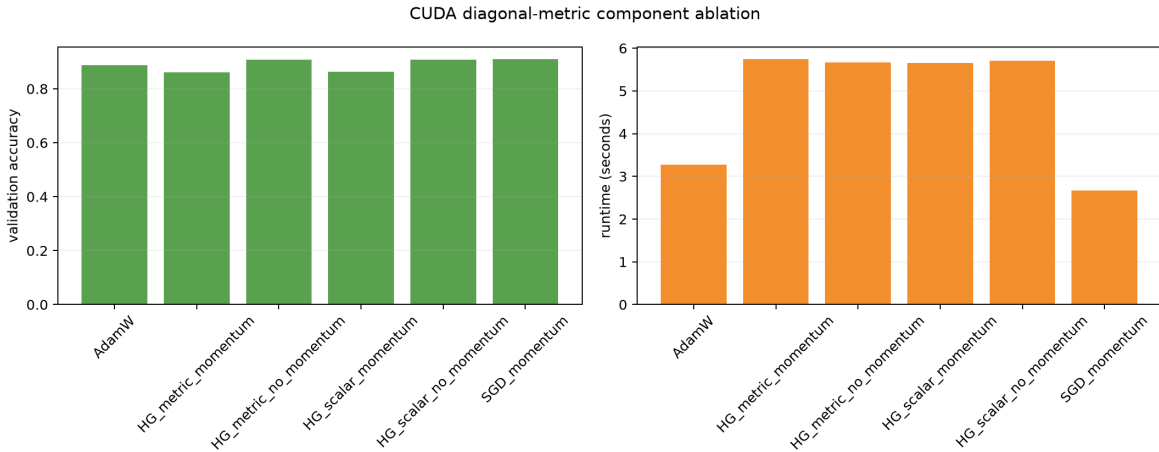


Figure 3: Accuracy and compute cost for the CUDA study.